A Technical Typology of Prediction Markets: Mechanics and Tradeoffs

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Prediction markets existed long before the current blockchain era, but it was with the emergence of Polymarket that they truly broke through into the mainstream.

Since its launch in 2020, @Polymarket's on-chain model demonstrated the power of combining a decentralized infrastructure with an accessible user experience, attracting significant trading volume for high-profile events like the U.S elections.

Its success served as a powerful proof that not only showcased the potentials for decentralized forecasting but also paved the way for a new wave of innovation and competition within the prediction market space.

Not all prediction markets are created equal.

The rise of Polymarket, also brought about the diverse and complex design choices that powers prediction platforms..

While they all share a common goal of aggregating collective intelligence, different platforms employ different approaches.

They differ dramatically in how they function, from the underlying technology that powers them to the mechanisms that determine a market's "truth.”

This article aims to provide a comprehensive typology of prediction markets,

The typology draws from real projects, ranging from established platforms like @Kalshi and Polymarket to emerging ones like @trylimitless , @xodotmarket and @MyriadMarkets.

The goal is to spot the technical choices projects make and their implications so as to help users and developers navigate the ecosystem's diversity, including the risks and potentials of prediction markets.

Let's begin

#1: Outcome Resolution

In prediction markets, Outcome resolution determines how a market's result is verified and settled which is crucial for trust and finality.

Projects differ in this class as they either rely on external data sources, human judgment, or automated processes, each introducing unique technical challenges. While new and upcoming projects often experiment with hybrid approaches to balance speed and reliability.

Oracle-Based Resolution

These Projects use decentralized or centralized oracles to fetch and verify off-chain data (e.g., results, prices, scores) for on-chain settlement and resolution.

Examples:

> **Polymarket**: A hybrid prediction market using @UMAprotocol optimistic oracles for arbitration and dispute resolution
>
> **Myriad:** An on-chain prediction market that employs oracles to submit proofs and resolve disputes,

Limitless: On Base, it uses oracle price feeds provided by @PythNetwork, used for resolving price-action markets. It also employs custom resolutions, particularly for user-generated markets

OpinionLabs (Upcoming): Building on Monad, @opinionlabsxyz is Primarily Oracle-based but leveraging on decentralized oracle networks to fetch external data to resolve market outcomes.

Human-Reviewed Resolution

Here, Outcomes are decided by community votes, expert panels, or special administrators within the platform. This resolution is suitable for subjective or ambiguous events and most times also depends on external oracle feeds.

Human review excels in nuances but sacrifices speed and risks centralization making it less scalable for high-volume, objective events compared to oracles.

Upcoming projects may use this in their early days and for community building but will likely face challenges in scaling...using a hybrid method is the best.

Examples:

Kalshi: A regulated, prediction platform with human oversight under CFTC guidelines. Resolutions are monitored for fairness, ensuring compliance. Kalshi has a dedicated markets team that thoroughly reviews markets and determines outcomes when the resolution criteria have been met.

Augur: A very early Ethereum-based project with community reporters for resolutions,

XO Market (Upcoming): While it is primarily permissionless, The resolution process starts with an AIAgent, known as MODRA (built by @xodotmarket) which resolves straightforward markets automatically. If the market outcome is disputed, the case escalates to a court of human jurors.... So it incorporates human review for ambiguous settlements.

Automated/Algorithmic Feeds

In this category markets are resolved automatically by an algorithm that directly processes data from a designated, pre-defined feed or data source.

The resolution is deterministic based on the input data, and human intervention is generally absent unless there's a dispute or error in the feed itself.

The problem here is Automation prioritizes efficiency over flexibility, trading the ability to handle ambiguity for faster payouts and resolution. It is suitable for crypto or price markets but not best for subjective topics like politics.

Examples

Kalshi: This is a prime example. their system is designed for automated settlement based on the data provided by their official sources which are basically preselected feeds.

Polymarket: Also relies on automated feeds, from integrated oracles that provide data from various sources (e.g., news APIs for event outcomes) powered by @chainlink. The resolution is triggered once the specified data point is met.

Limitless: As mentioned, Limitless uses Pyth, a decentralized oracle that aggregates data from numerous sources. The market resolution is driven by these automated feeds.

SWOT Analysis

#2: Liquidity Model

Liquidity determines how easily users can enter/exit positions without price slippage.

The liquidity model a prediction platform uses affects trading efficiency and capital requirements.

High liquidity ensures that individual trades have minimal price impact. This allows the market price to move smoothly and predictably as new information arrives, accurately reflecting the crowd's evolving forecast.

Low liquidity can cause prices to move erratically, as a single large order can cause disproportionate price swings.

New projects often innovate here to bootstrap liquidity in early stages.

Automated Market Maker (AMM)

This method depends on algorithms powered by mathematical formulas to provide constant liquidity without matching buyers/sellers. Users trade against a liquidity pool, and the price of the assets in the pool adjusts continuously.

AMMs ensure liquidity is always ready in new markets but trade precision for simplicity, unlike order books that offer better prices at scale. One popular example is the Logarithmic Market Scoring Rule (LMSR), which provides continuous liquidity and incentivizes informed trading.

Prediction markets often start as AMMs then migrate later like Polymarket did

Examples

- **XO Market**: Uses LS-LMSR, a type of AMM specifically optimized for prediction markets.
- **Augur:** Uses AMMs (specifically order books that mimic AMM behavior) to ensure liquidity
- **Myriad:** Myriad utilizes a vAMM to provide deep, constant liquidity for its prediction markets.

Order Book/Centralized Limit Order Book (CLOB)

This is the traditional model used in financial exchanges. Buyers and sellers place limit orders (orders to buy/sell at a specific price) or market orders

(orders to buy/sell immediately at the best available price). The platform then matches these orders.

This model Requires active users or professional market makers to maintain deep liquidity. Without sufficient activity, markets can be illiquid and difficult to trade.

CLOBs provide better capital efficiency for pros but require more initial liquidity than AMMs, making them harder to launch for small projects.

Examples:

> **Polymarket:** Polymarket initially used an AMM but now operates as an order book where users can place limit orders to buy or sell "outcome shares" at specific prices. The order book facilitates price discovery based on supply and demand.
>
> **Limitless**: uses CLOB for trading outcome shares, fitting its social, creator-configured markets
>
> **SX Network:** Operates its own custom EVM chain with order matching for on-chain settlements,

Market maker / House based

A central entity (the "house" or platform operator) acts as the counterparty to all bets. It sets the odds (prices) and takes positions opposite to users.

The house always provides liquidity, so users can always place bets. This is very common among traditional sportsbooks.

Example:

@DivvyBet: A Decentralized sportsbooks on Solana that allows anyone to be the house

SWOT Analysis

Hybrid models

This model combines elements from two or more of the mentioned models to leverage their strengths and mitigate their weaknesses... They aim for optimal liquidity, price discovery, and capital efficiency.

Kalshi uses a hybrid liquidity model that is centered around a CLOB but is heavily supported by both an internal trading arm and professional, third-party market makers.

This model ensures deep liquidity across its markets, a crucial feature for any regulated financial exchange.

Kalshi's approach most closely aligns with the Hybrid model, but with a strong emphasis on the Market Maker/House-Based and Order Book-Based components.

It differs significantly from AMM-based models by not relying solely on algorithms and liquidity pools provided by the general user base.

#3: Event Creation Mechanisms

Prediction markets can also be classified by their event creation mechanism, which determines how markets are proposed, vetted, and launched.

This process is a key differentiator between platforms as it impacts their market diversity, quality, and overall philosophy (centralized vs decentralized).

Platform-curated (Permissioned)

The platform operator or a central team has full control over which events are offered to users.

The platform acts as a gatekeeper, and users can only trade on events selected and launched by the platform.

This model is often preferred by regulated platforms, as it allows for greater control over event legality and compliance.

Example:

> **Kalshi:** Kalshi curates all of its markets, which are vetted for clarity and regulatory compliance.
>
> **PredictIt:** Historically, this platform curated all political markets to ensure they adhered to its regulatory structure.

User-generated (Permissionless)

Here, any user can propose and create their own market, as long as they follow the platform's rules and deposit the necessary collateral.

This model is often used by decentralized platforms running on Blockchains.

Permissionless platforms often need additional mechanisms, like curation markets to handle low-quality or fraudulent events.

Example:

> **Augur:** Was one of the pioneers of permissionless market creation, allowing anyone to propose a market.
>
> **XO Market**: Leverages user-generated, permissionless market creation.
>
> **Zeitgeist**: Allows permissionless market creation on its Polkadot parachain.

Community-vetted

A hybrid approach where users can propose markets, but the final decision to launch them is made by a community-based governance process or a multi-stage voting system.

This provides a balance between free-form user creation and the quality control of a curated platform.

Example:

Polymarket: Users can suggest, discuss, and review new markets through a channel, influencing which ones the Polymarket team ultimately launches. This acts as a form of centralized vetting process informed by community sentiment.

SWOT Analysis

#4: Market Type

This refers to the different structures or formats that prediction markets use for expressing probabilities and facilitating trading on outcomes.

The market type determines how probabilities are represented, how contracts are designed, and how payouts are structured. A single platform can offer various market types.

Binary markets

These are the simplest type of prediction market. There are only two possible outcomes, typically framed as a "Yes" or "No" question.

The market resolves to either 0% or 100% probability.

The price of a "Yes" contract directly represents the market's perceived probability of that outcome (e.g., a "Yes" contract trading at $0.70 implies a 70% chance).

Examples: Kalshi, polymarket, XO...

Multi-outcome markets

These markets involve more than two possible discrete outcomes, but only one can be true. For example, "Which candidate will win the election?" where the options are Candidate A, Candidate B, or Candidate C.

This type often benefits from AMM designs (like LMSR) that can manage the relationships between the probabilities of multiple outcomes.

Here is a good example: "Which team will win the World Cup?" with outcomes for each of the top contenders.

> If there are eight teams left in the tournament, eight separate "outcome tokens" would be created.
>
> Each token's price would fluctuate based on trades, with the price reflecting the market's probability of that team winning.
>
> The market aggregates these varying probabilities to predict the most likely winner.

A multi-outcome market on Polymarket to predict which team wins the EPL

Scalar Market

Scalar markets involve predicting a numerical value or range (e.g., "What will Bitcoin's price be on Date X?"), with payouts scaled linearly or in bands based on how close the prediction is to the actual outcome.

Technically, this uses quadratic scoring rules, which are resolved via quantifiable data feeds (e.g., price-based oracles), enabling precise, data-driven settlements.

Take for instance a market asking for the closing price of a stock (e.g., between $100 and $150).

If a participant believes it will close at $125, they would take a position reflecting that belief. If the stock closes at $120, they would receive a payout proportional to how close $120 is to their predicted $125, demonstrating the reward for precision.

Example: @Trepa_io (in beta) , a prediction platform focused on precision and accuracy rather than binary outcomes.

@trepa_io

It basically reward traders based on how close their forecast is close to the

real outcome.

SWOT Analysis

SWOT analysis of market types used in Prediction Markets

#5: Underlying Infrastructure

The underlying infrastructure of a prediction market significantly impacts its performance, security, and the user experience it provides.

This infrastructure determines how transactions are processed, data is stored, and the overall level of trust users can place in the system.

Off-chain Infrastructure

The core operations of the prediction market, including market creation, order matching, user account management, and often fund custody, run on servers controlled by a single entity.

The use of a public blockchain might be minimal or absent, primarily serving as a public ledger for auditing in specific cases, if at all.

Example: Kalshi, currently planning to expand its onchain presence.

On-chain Infrastructure

Here the fundamental components of the prediction market, such as market creation, trading, outcome resolution logic, and fund management are executed and recorded via smart contracts on a blockchain.

User funds remain in their own wallets (non-custodial) or controlled by smart contracts, rather than held by a central entity.

Examples: polymarket, Xo market, limitless, myriad

SWOT Analysis

SWOT analysis based on underlying infrastructure

Hybrid prediction markets combine aspects of both on-chain and off-chain systems.

These models aim to leverage the benefits of each approach, such as the security and transparency of on-chain settlement with the speed and efficiency of off-chain processing.

For instance, @Theclearingco is developing regulated on-chain prediction platforms that integrate on-chain efficiency with a clear compliance framework.

Final thoughts

One of my most interesting takeaways from this classification is the broad spectrum of decentralization in play.

On one end, we have centralized players like @Kalshi , which prioritize regulatory compliance and deep institutional liquidity.

On the other end are fully on-chain protocols using cryptoeconomic incentives and governance to achieve censorship resistance and trustlessness.

Hybrid platforms also demonstrate an ability to balance the strengths of both worlds, leveraging off-chain speed for order matching while retaining on-chain security for settlement.

Also, the quest for a perfect liquidity model is also far from over.

While CLOBs dominate, AMMs cannot be overlooked. Projects like XO Market

push this further with customized AMM algorithms (LS-LMSR) that optimize for capital efficiency and market depth.

You can agree that the success of a prediction market often depends on its ability to solve the liquidity problem, and the race to find the most efficient and scalable model is a primary driver of innovation in this space.

Ultimately, there is no single "best" prediction market.

The ideal design depends entirely on the market's target audience, use case, and tolerance for trade-offs. A well regulated prediction market serves a different purpose than a censorship-resistant one.

For example, Platforms like Trepa focus on incentivizing precise prediction while other platforms provide binary predictions.

I believe we can expect to see an even richer pattern of design choices as the tech and ecosystem expands.

In conclusion, the next chapter will be written by platforms that can most effectively balance the core components of liquidity, resolution, and user participation to create markets that are genuinely useful for forecasting the future.

An that is wrap!

Tagging relevant folks

@guyukyukgu, @superteamkorea, @factcheckdotfun, @marvelloudefi_, @0xnairolf, @johnwang, @aulijk, @defi0xjeff, @_drNeyo, @0xcheeezyyy, @0xtone, @_cmger, @netroverthq, @pinnaclecrypt, @netroverthq, @iam_lerik, @defi

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